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/* I declare that this work is my own.  
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How can computer graphics renderings of statues be made to look more realistic?

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Abstract

This report explores how visual realism can be achieved in the real-time rendering of statues. It compares an empirical lighting approach based on the Phong reflection model with a Physically Based Rendering (PBR) pipeline. By evaluating the use of multi-channel texture mapping, including albedo, roughness, metallic, and normal maps, the study demonstrates how modelling light interaction with micro-scale surface geometry contributes to the realistic appearance of materials such as aged stone, weathered wood, and brushed metal. The results indicate that realism is not determined solely by high-resolution textures, but also by the use of physically plausible models of light-material interaction under real-time rendering constraints.

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1 Introduction

Achieving a sense of realism in visual representation is a primary objective of 3D computer graphics, particularly when applied to artefacts such as statues that exhibit complex surface characteristics. These objects present unique challenges for light–material interaction. This report addresses the question of how computer graphics renderings of statue-like objects can be made to look more realistic through texture mapping within a real-time rasterisation pipeline.

The report is structured to guide the reader from empirical approximations to physically based simulation. Section 2 introduces the fundamentals of texture mapping and discusses the importance of appropriate texture resolution and filtering for practical implementation. Sections 3 and 4 serve as the main “Contrast and Appraisal” components of the assignment, comparing the Phong-based approach introduced in Assignment 1 with a Physically Based Rendering (PBR) approach. This structure aims to illustrate the visual “leap” from appearances that are merely plausible to those that are physically grounded. Finally, Section 5 presents a visual comparison between real-world materials and their digital counterparts using image triplets.

The discussion is based on three image triplets, each consisting of a real-world photograph, a Phong-based rendering, and a PBR-based rendering, enabling direct visual and analytical comparison.

2 Fundamentals of Texture Mapping

Realistic rendering of statues depends to a great extent on how surface detail is represented and sampled during rasterisation. Texture mapping allows fine-scale material detail to be added without increasing geometric complexity; however, the resulting visual quality depends strongly on how texture data is resolved and filtered under different viewing conditions. In the context of statue rendering, where surfaces are typically large, vertically oriented, and viewed from a range of distances, these considerations become particularly significant.

2.1 Texture Resolution and Mipmapping

High-resolution textures are required to capture the fine surface detail of real-world statue materials, such as the granular structure of stone and the fibrous grain of wood. However, when such textures are viewed at a lower screen-space resolution, aliasing artefacts may occur, often appearing as flickering or shimmering when the camera moves through the scene.

Mipmapping addresses this issue by enabling level-of-detail selection within the texture itself [3]. For tall Menhir-like statues that are commonly viewed both up close and from a distance, mipmapping helps maintain visual stability across different viewing distances while preserving material consistency at multiple scales. By reducing distracting aliasing artefacts, mipmapping contributes directly to the perception of realism, as visual instability would otherwise undermine the physical plausibility of the rendered surface.

2.2 Texture Filtering and Anisotropic Sampling

Texture filtering also affects how surface detail is perceived when surfaces are viewed at shallow angles. Simple filtering methods can result in excessive blurring on steeply inclined statue surfaces, such as vertical faces or edges extending away from the camera.

Anisotropic filtering mitigates this issue by adapting the texture sampling footprint according to the viewing direction [3]. For statue rendering, this is particularly relevant, as large planar regions are often viewed at grazing angles. By preserving high-frequency detail under these conditions, anisotropic filtering helps retain consistent material appearance and supports the visual continuity required for realistic real-time rendering.

3 Lighting Models for Real-Time Statue Rendering

The realism of a rendered statue depends less on its geometric appearance alone and more on how light interacts with the surface material. In a real-time rasterisation pipeline, this interaction is governed by the chosen lighting model. This section introduces the empirical Phong reflection model presented in Assignment 1 and provides a conceptual comparison with the physically motivated approach used in Physically Based Rendering (PBR). It establishes the theoretical basis required for the visual comparisons presented later in this report.

3.1 Phong Reflection Model

The Phong reflection model is a widely used empirical shading approach designed to approximate light reflection using a small set of intuitive parameters [1]. The reflected intensity I at a surface point is computed as:

$$I = k_a I_a + \sum_{i=1}^n [k_d(\mathbf{L}_i \cdot \mathbf{N}) + k_s(\mathbf{R}_i \cdot \mathbf{V})^\alpha] I_{i,light} \quad (1)$$

where k_a , k_d , and k_s control the ambient, diffuse, and specular components respectively, and α represents the shininess of the material. The model is computationally inexpensive and has historically been well suited to real-time applications.

However, the Phong model is not physically based. In particular, it does not enforce energy conservation, allowing diffuse and specular components to be independently scaled beyond physically plausible limits [3]. Furthermore, the specular term is not intrinsically linked to the material's base colour or physical composition. As a result, different materials often require extensive manual tuning to achieve visually distinct appearances.

For statue rendering, these limitations are especially apparent. Materials such as stone, wood, and metal exhibit distinct visual characteristics due to differences in micro-scale surface structure and reflectance properties. While the application of colour textures can introduce surface variation, the Phong lighting response remains largely insensitive to such variation. This limits the extent to which realistic material appearance can be achieved using this approach.

4 Physically Based Rendering

Physically Based Rendering (PBR) addresses the limitations of empirical lighting models by approximating light-surface interaction using physically motivated principles. Rather than relying on artist-defined coefficients, PBR derives material appearance from measurable or physically interpretable properties, resulting in greater consistency under changing lighting conditions.

4.1 Core Principles of PBR

Modern PBR pipelines are conceptually grounded in the Rendering Equation and commonly employ microfacet theory to describe surface reflectance [2]. As emphasised in the lecture material [3], two principles underpin most real-time PBR implementations:

- **Energy Conservation:** The total amount of reflected light energy does not exceed the amount of incident light energy.
- **Microfacet Surface Representation:** The surface is modelled as a distribution of microscopic facets, where the orientation of each microfacet contributes to the appearance of specular reflection.

These principles are realised in practice through the use of multiple texture maps, each controlling a specific material property. The albedo map defines the base colour independent of lighting, while the roughness map controls the distribution of microfacets, influencing the sharpness or blurriness of specular highlights. The metallic map distinguishes conductive from non-conductive materials, ensuring a physically plausible separation between diffuse and specular reflection. Additional maps, such as normal and ambient occlusion, encode fine-scale surface detail and localised light attenuation. Modern real-time PBR pipelines typically employ simplified microfacet-based shading models derived from these principles, as demonstrated in production systems such as Disney’s physically based shading framework [5].

4.2 PBR Material Workflow and Data Complexity

In this project, PBR was carried out on existing PBR material sets sourced from online repositories which had permissive licences. Unlike the Phong-based approach, which relies on a single photographic texture applied directly to the statue surface, the PBR workflow employs multiple texture maps integrated within a node-based material system.

Figures 1–3 show the actual material node graphs used for the stone, wood, and metal statues respectively. Each graph represents a complete PBR material configuration, incorporating albedo, roughness, metallic, and normal maps, with additional nodes used for value adjustment and coordinate control as required by the material.



Figure 1: PBR material node graph used for the stone statue.

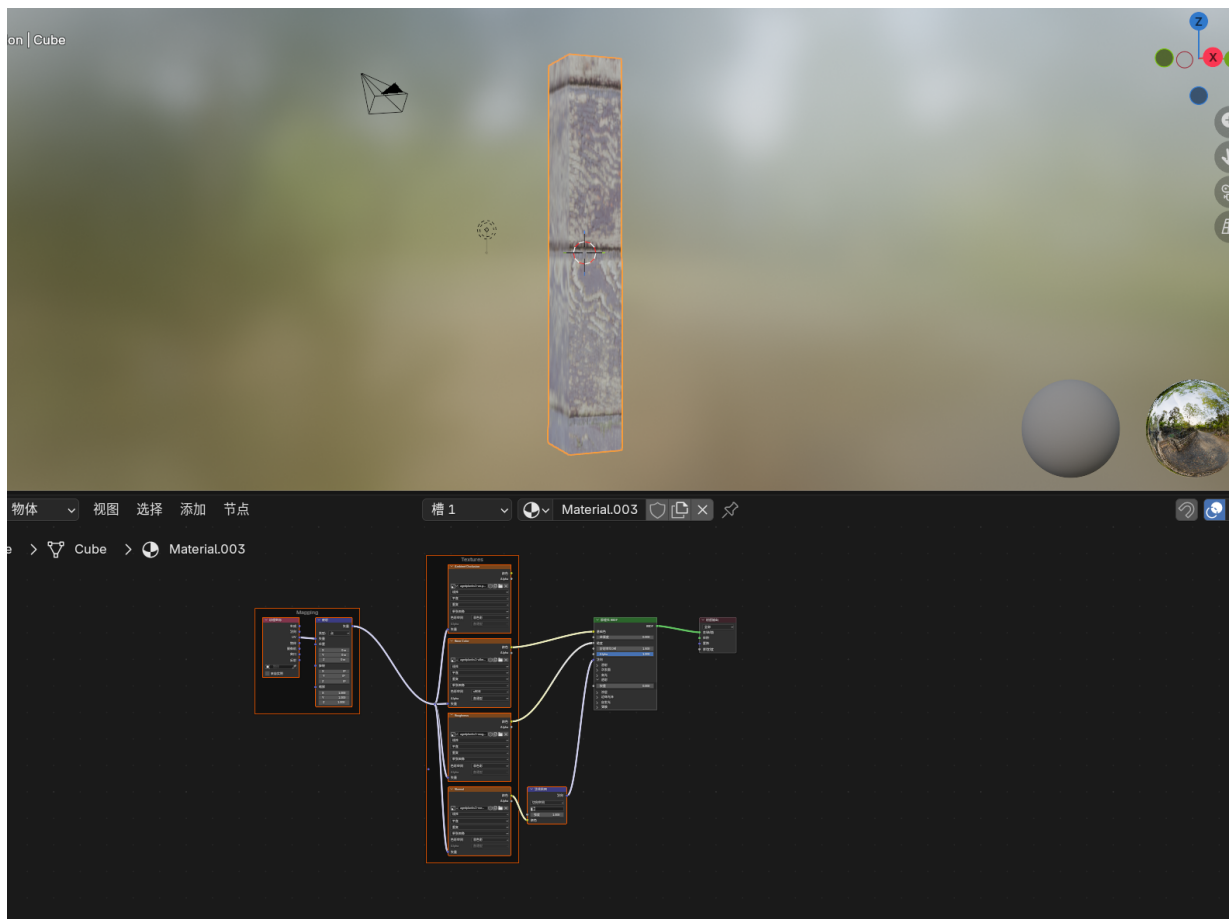


Figure 2: PBR material node graph used for the weathered wood statue.



Figure 3: PBR material node graph used for the metal statue.

Compared to the Phong workflow, the PBR material setup demonstrates a substantially higher data and configuration complexity. Not manually tuning lighting coefficients but the result of material behaviour is instead the result of the interaction of several texture-driven parameters. Changing the surface roughness or metallic property automatically leads to a change in reflectance, which is still reasonable energy distribution.

This difference in workflow draws our attention to a trade-off in real-time rendering. Phong’s approach offers ease and quickset up, yet it gives less control over physically viable surface behaviour. Although the PBR method is more demanding for texture data and material configuration, it can make the appearance of the rendering result closer to the real-world photograph reference. And these differences are an important basis for the visual comparisons given in Section 5.

4.3 Practical Implications for Statue Rendering

The increased complexity of the PBR workflow represents a trade-off between setup effort and visual fidelity. For statue rendering, where materials often exhibit subtle surface irregularities and weathering effects, this trade-off is justified. PBR allows these irregularities to influence the lighting response directly, improving consistency with photographic reference material.

PBR does not promise to be the most real. Some real world effects like the anisotropic reflection on a brushed metal or large scale geometric erosion will still be approximated or absent in real time pipelines. But in rasterised, PBR provides a more physically coherent basis than empirical lighting

models. Section 5 in the form of visual comparison with actual photos are observed directly: differences are examined: Section 5: Differences are examined visually by direct comparison with real-world photos.

The specific PBR material sets were sourced from FreePBR, with the scratched metal, rock mortar, and aged wood planks materials adapted for this project [6, 7, 8].

5 Visual Evidence and Comparative Analysis

This section assess the effectiveness of different renderings through direct visual comparison. A triplet of images is displayed for each material: the first is a photograph of a real-world object, the second is a rendering result based on the Phong model, and the third is a PBR result. These triplets serve as the main evidence for judging the impact of texture-driven lighting models on the real-world effect of statue materials.

5.1 Material Comparison 1: Stone

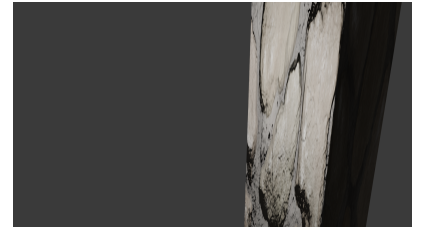
Stone statues are predominantly characterised by diffuse reflectance, surface porosity, and irregular micro-scale geometry resulting from natural erosion. These properties make stone an effective material for evaluating how different lighting models respond to subtle surface variation.



(a) Real photograph



(b) Phong-based rendering



(c) PBR-based rendering

Figure 4: Triplet 1: Comparison of stone material appearance.

In the Phong based rendering colour variations derived from the photographic texture give a basic idea of stone. However, when the light shines on it, the surface can be seen as smooth with the naked eye, and there are fine pits and depressions seen in the picture do not affect the Lighting much. This is due to the fact that the Phong model does not consider the affectation of the micro-surface structure on the scattering of light.

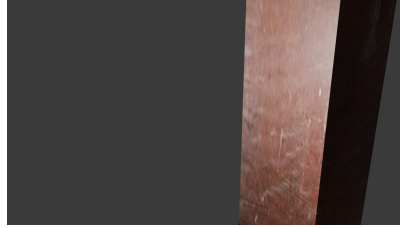
PBR rendering shows a more similar representation of the actual material. The combination of roughness map and normal map can generate roughness irregularities on the surface of the model, which may have slightly softer highlights and more detailed shadows that are similar to the reference. The comparison shows that for materials which are predominantly diffuse reflection, realism does not rely so much on colour accuracy as it does on the control of light interactions.

5.2 Material Comparison 2: Wood

Weathered wood has a clear direction of the wood grain, surface cracks and uneven roughness caused by long-term exposure to the environment. These characteristics present a difficulty for empirical lighting models because surface structure has to meaningfully affect the reflectance to work.



(a) Real photograph



(b) Phong-based rendering



(c) PBR-based rendering

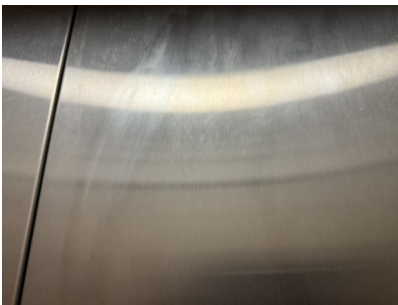
Figure 5: Triplet 2: Comparison of weathered wood material appearance.

Phong Rendering photography texture, general wood grain pattern but it looks un-deep on structural features such as cracks, fiber direction. These information is expressed mainly through color changes and does not make the lighting behave differently, so it becomes flat in appearance.

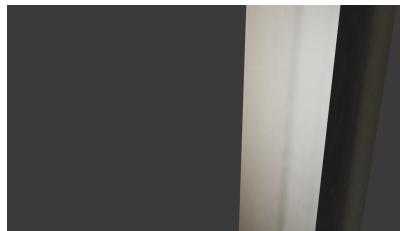
PBR renders more physically correct on the material. Different roughness values plus normal mapping so cracks, grain direction would affect where the highlights went and the amount of shading. This creates a stronger feeling of the form of structure, which is closer to the real wood surface. Comparison shows that PBR is able to make materials real by embedding surface structures into the lighting responses rather than on albedos alone.

5.3 Material Comparison 3: Metal

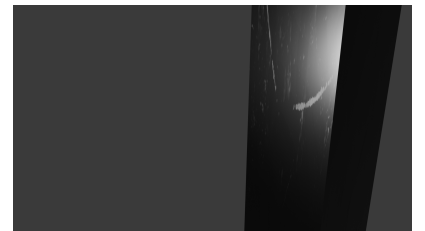
Metallic materials is defined by reflect, not surface color. Real metals have limited diffused reflection and most of their appearance is due to the specular response, so they are more sensitive to the choice of light model.



(a) Real photograph



(b) Phong-based rendering



(c) PBR-based rendering

Figure 6: Triplet 3: Comparison of metal material appearance.

In the Phong-based rendering the appearance of a metal is approximated by a strong specular component. This does make the highlights very noticeable but at the expense of not influencing the reflectance behaviour by the material’s actual physical properties and thus the material could take on a plastic like look depending on the lighting presented.

PBR rendering can give a more physical coherence representation. By making use of a metallic map, the material properly shifts between a diffuse type and a specular type. While still all, and won’t reproduce any of the anisotropic reflections of a brushed metal, but for sure more realistic. This comparison shines both the advantage and the left shortage of PBR under real-time rendering conditions.

6 Representing Time and Weathering

While material representation is important for realism, many real-world statues derive much of what we see from the effect of time. Environmental exposure gradually produces eroded, stained, blackened, biological growth and so on, thus changing the relationship of the material and light interaction. Representing such temporal effects is an extra burden to real time rendering pipeline’s capability of rendering.

In contemporary texturing pipelines, such aging effects are frequently emulated via texture blending as opposed to direct geometry alteration. For example, erosion of stone surface can simulate with roughness varying and detail in normal map; Discoloration on the surface may be inserted into the albedo texture directly. And similar weathered wood we might see the same kind of surface roughing up and the same kind of reflectance caused by cracks and fibres exposed on the surface, which again we could encode without increasing the mesh itself.

Mask based blending is often used to create complex temporality such as Moss Growth / corrosion. A secondary material layer is selectively applied using noise functions or curvature based masks so weathering will accumulate in crevices and shadowed regions where moisture would realistically persist [4]. Though these are still approximations, they offer a plausible visual way of adding some time-varying thing under real-time conditions.

Although there have been these advancements, there are still some restrictions. Large-scale erosion that changes the outline shape can not be accurately presented without modifying the geometry, and it is also difficult to express the anisotropic wear pattern with the standard isotropic lighting model. Therefore, although PBR and texture - driven ageing techniques can significantly improve the realism of the visual effects, they still cannot fully reproduce geometric evolution or physically simulated degradation. These defects show the compromise of visual fidelity and performance in real-time rendering technology itself.

7 Conclusion

This report has examined how computer graphics renderings of statues can be made more realistic within a real-time rasterisation framework. By contrasting an empirical lighting approach based on the Phong reflection model with a Physically Based Rendering (PBR) pipeline, the study demonstrates that realism does not arise solely from high-resolution textures, but from physically coherent modelling of light-material interaction.

Visual comparisons across three material triplets (stone, wood, and metal) show that Phong shading is capable of producing a basic visual appearance through colour variation. However, it struggles to convey structural depth and material-specific reflectance behaviour. In contrast, PBR enables texture data such as roughness, metallicity, and normal information to directly influence shading behaviour, resulting in renderings that more closely resemble photographic reference materials.

The analysis also highlights that PBR is not a complete solution. Certain real-world effects, such as anisotropic reflection and long-term material degradation, remain difficult to reproduce accurately under real-time constraints. Nevertheless, when combined with appropriate texture mapping, filtering

techniques, and ageing approximations, PBR provides a robust and physically grounded approach for improving the realism of statue rendering.

In conclusion, realistic computer graphics renderings of statues are achieved by integrating physically based lighting models with carefully constructed texture data, while acknowledging the inherent trade-offs between visual fidelity, computational complexity, and performance in real-time rendering systems.

References

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- [2] Cook, R. L., and Torrance, K. E. (1982). A reflectance model for computer graphics. *ACM Transactions on Graphics*.
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A Appendix A: Practical Implementation

A.1 A.1 Photography Details

All photographic reference images used in this report were captured by the author using a smartphone camera (Apple iPhone 16 Pro Max). Photographs were taken in a residential outdoor environment in China under natural daylight conditions. No artificial lighting setups were used, and all images were captured handheld.

The camera was operated in the standard automatic photo mode, allowing the device to select exposure parameters such as aperture, shutter speed, and ISO automatically. This approach was chosen to reflect typical real-world observation conditions rather than a controlled studio environment, ensuring that the captured images represent how statue-like materials are commonly encountered in everyday settings.

Three primary photographs were selected for inclusion in the main body of the report, corresponding to the three materials analysed: stone, wood, and metal. Each photograph was chosen to emphasise surface characteristics relevant to realistic rendering, including colour variation, surface roughness, structural irregularities, and reflectance behaviour. These images form the photographic reference component of the image triplets discussed in Section 5.

In addition to the primary reference photographs, supplementary environmental images were captured at each location. These images provide contextual views of the materials in situ, illustrating how surface appearance varies with viewing angle and natural lighting direction. While these environmental photographs are not included in the main comparative analysis, they support the interpretation of material properties and are presented here for completeness.



Figure 7: Environmental photograph showing the stone material in its surrounding context under natural lighting conditions.

The stone material photographs highlight granular surface structure, colour non-uniformity, and rounded edges resulting from natural weathering. These features informed the evaluation of diffuse-dominated reflectance behaviour and surface roughness variation in the rendered stone statue.



Figure 8: Environmental photograph illustrating the weathered wood material and its exposure to outdoor conditions.

The wood photographs focus on directional grain, surface cracking, and ageing effects caused by prolonged environmental exposure. These characteristics were particularly relevant when assessing how effectively different rendering approaches represent structural depth and surface irregularity.



Figure 9: Environmental photograph showing the metal surface under natural lighting, highlighting reflective behaviour.

The metal photographs capture reflective behaviour and surface finish under natural illumination. These images were used to assess how effectively Phong-based and Physically-Based Rendering approaches reproduce metal-specific reflectance characteristics, as discussed in Section 5.3.

A.2 Assignment 1 Adjustments and Phong-Based Rendering

The Phong-based renderings used in this report were produced using Blender. The Phong reflection model was configured within Blender's material system to approximate the empirical lighting approach studied in Assignment 1.

To ensure that the comparison focused exclusively on statue material appearance, the scene was

simplified to contain a single Menhir-style statue under a basic lighting setup. Non-essential elements such as additional objects or complex lighting arrangements were omitted to maintain consistency across all renderings.

For each material, a single photographic texture captured by the author was applied directly to the statue geometry as a diffuse texture. No additional texture channels, such as normal, roughness, or metallic maps, were used in the Phong-based pipeline. Material appearance was therefore controlled primarily through the diffuse colour information contained in the photograph, alongside manually adjusted Phong lighting parameters.

This simplified setup reflects the inherent limitations of empirical lighting models in real-time rendering. While the photographic texture introduces surface colour variation, the lighting response remains largely uniform across the surface and insensitive to fine-scale structural detail. These constraints were intentionally preserved to provide a clear baseline against which the Physically-Based Rendering results could be evaluated in Section 5.

A.3 Physically Based Rendering Setup

Physically Based Rendering was implemented using Blender’s node-based material system. Rather than creating custom PBR textures from scratch, pre-existing material sets were sourced from online repositories providing physically based texture data under permissive (CC0) licences. These resources were selected to closely match the visual characteristics observed in the real-world photographs captured by the author.

For each statue material (stone, wood, and metal), a complete PBR material was constructed by integrating multiple texture channels, including albedo, roughness, metallic, and normal maps. Where required, additional nodes were used to adjust texture scaling, coordinate alignment, and value ranges to ensure appropriate correspondence with the Menhir geometry. No displacement mapping was applied, and all surface detail was represented through texture-driven shading.

Figures presented in Section 4.2 show the actual node graphs used for each material. These graphs illustrate the increased data and configuration complexity of the PBR workflow compared to the Phong-based approach. Rather than manually tuning lighting coefficients, material behaviour emerged from the interaction of physically meaningful texture parameters, resulting in more consistent and realistic surface appearance under the same lighting conditions.

The use of externally sourced PBR materials highlights an important practical consideration in real-time rendering. While the PBR approach requires greater data input and setup effort, it enables a closer alignment between rendered appearance and photographic reference material. All external texture resources used in this project are listed in the reference section, with licensing information provided where applicable.

B Appendix B: Generative AI Statement

Declare: I confirm that no content created by generative AI technologies has been presented as my own original work without critical review and modification. All analysis, interpretation of results, experimental decisions, and final conclusions presented in this report are the author’s own.

Signed: Yongjiang Liu

Date: 16 January 2026

Acknowledge: I acknowledge the use of the following generative AI systems for limited and specific support purposes:

- ChatGPT (OpenAI) was used to review the overall report structure, improve academic phrasing, and check clarity and coherence of written explanations.
- Google Gemini was used to assist with learning the practical use of Blender, including understanding material node connections and general workflow guidance.

Describe: Generative AI tools were used strictly as supplementary support and did not replace the author’s own academic work. ChatGPT was employed to review draft sections for structural consistency, suggest improvements in academic tone, and help ensure alignment with the assignment brief. All suggested text was critically evaluated, edited, and integrated by the author.

Google Gemini was used during the practical phase of the project to clarify how specific features of Blender operate, such as navigating the node-based material editor and connecting PBR texture maps. All decisions regarding rendering approaches, texture selection, material configuration, and visual analysis were made independently by the author based on course materials and experimental outcomes.

Evidence: Typical prompts included requests for feedback on report organisation (e.g., whether the section structure aligned with the assignment requirements) and practical questions related to Blender usage (e.g., how to connect albedo, roughness, metallic, and normal maps within the material editor). No generative AI system was used to generate rendered images, experimental results, or unedited analytical content included in this report.